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Effects of short inter-repetition rest periods on power output losses during the half squat exercise

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Abstract.

BACKGROUND: Cluster training is being increasingly used to develop muscular power.

OBJECTIVE: To determine the effects of short inter-repetition rest (IRR) periods on the capacity to maintain maximal levels of power output.

METHODS: In a first session, 16 active-duty soldiers performed a progressive loading test to establish the load linked to maximal power (optimal load, OL), and the half squat 1-repetition maximum. In Session 2, six individual sets of repetitions performed to failure (or a maximum of 20 repetitions) were conducted using the loads OL, low (LL, 15% below OL), and high (HL, 15% above OL) as quickly as possible. For each load, participants performed one set without rest between repetitions (CR, continuous repetition protocol), and another set with 6 s of rest between repetitions (IRR protocol).

RESULTS: The number of repetitions participants performed before exceeding a power loss threshold of 15% were higher in the IRR versus the CR protocol by 218% (11 vs. 35), 86% (7 vs. 13), and 175% (4 vs. 11) for LL, OL, and HL, respectively.

CONCLUSIONS: A 6 s interval between repetitions is sufficient to induce partial recovery in participants, and could therefore improve muscle power output.

Keywords: Muscle fatigue, cluster training, load-power relationship, power training

1. Introduction

Maximizing speed and power during each repetition should be a priority for athletes attempting to develop muscular power [1]. Several authors have argued a fast rate of force development [2], and movement velocity [1] are key to elicit neuromuscular adaptations. In addition, when attempting to maximize power output, fatigue must be minimized since it will re-

duce power [3]. A simple way of avoiding fatigue is to execute few repetitions per set [2,4]. Several studies have addressed the optimal number of repetitions required to ensure maximal power outputs for different exercises and workloads [5,6]. For example, Baker and Newton [5] suggested athletes pursuing maximal power should limit their repetitions to 2 to 5 when using loads of 35 to 45% of their one-repetition maximum (1RM) in the bench press throw and jump squat. Information on the ideal number of repetitions needed during power training is useful for coaches and athletes.

Recently, a training modality known as cluster-set (CS) training has started to attract the attention of re-

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searchers as an alternative method of maintaining maximal levels of velocity and, thus, power output [7–10]. In CS training, short rest periods (15–30 s) are introduced between each rep (inter-repetition rest, IRR) or cluster of repetitions (intra-set rest, ISR). The goal of such rest periods is to minimize neuromuscular fatigue [9], and thus improve force, velocity and, power variables compared with traditional continuous training [11]. Lawton, Cronin and Lindsell [9] reported increased power outputs (21.6%–25.1%) during the bench press in response to three different CS configurations (6×1 rep, IRR 20 s; 3×2 repetitions, ISR 50 s; 2×3 repetitions, ISR 100 s) compared with the execution of traditional sets. Hardee et al. [8] observed greater losses in peak power output during six power clean repetitions at 80% 1RM performed as continuous sets (15.7%) versus the incorporation of rest periods of 20 s (5.5%) or 40 s (3.3%) between repetitions. Hansen, Cronin and Newton [4] similarly detected a significant drop in power output ($p < 0.05$) from repetitions 4–5 during the traditional set configuration compared to the power output response to three different CS configurations (6×1 rep, IRR 12 s; 3×2 repetitions, ISR 30 s; 2×3 repetitions, ISR 60 s) during the jump squat with a 40 kg weight ($\sim 20\%$ 1RM). Collectively, these findings support the CS protocol for several power training exercises.

An ability to develop high muscular power is essential for sports in which the time available to apply a force is limited (e.g., sprinting, throwing, jumping, etc.) [12]. Power is the product of muscular force and velocity (power = force $\times$ velocity), and maximal power output occurs at intermediate levels of maximal force and velocity [13,14]. Thus, the ability to generate power depends dramatically on the load applied to the athlete [12]. Optimal load (OL) is defined as the load that maximizes power output, and this load has shown to elicit the greatest adaptations in muscular power [15]. However, during power based training, athletes usually lift both heavy and light loads to improve force and velocity, respectively [12,13].

The behaviour of muscular power during training sets using these three load/velocity combinations (low, optimal, and heavy) has not been clearly established. Therefore, the objective of the present study was to determine the effects of introducing a short IRR period (6 s) on the capacity to maintain a given power output across sets of half squats performed at workloads commonly used to improve lower limb muscular power.

2. Materials and methods

2.1. Participants

Sixteen Spanish male active-duty soldiers (age: 33.7 ± 4.5 years, weight: 77.6 ± 7.7 kg, height: 175.3 ± 8.0 cm, half squat 1RM relative to body mass: 1.96 ± 0.24 kg $\cdot$ kg⁻¹) volunteered to participate in this study. The inclusion criteria for participation were: (a) an ability to lift at least 1.5 times their body weight during the half squat exercise; (b) at least six months of experience in half squat training; and (c) the non-use of drugs or dietary supplements that could affect physical performance. All participants were informed of the benefits and risks of the investigation prior to signing an informed consent form in accordance with the University of Granada Institutional Review Board and The Code of Ethics of the World Medical Association (Declaration of Helsinki).

2.2. Study design

Each participant undertook two testing sessions, 48–72 hours apart. In Session 1, the participant's load-power relationship and maximal dynamic strength (1RM) were determined in the half squat exercise. The OL was calculated individually by the polynomial equation method using load (kg) as the independent variable and mean power (W) as the dependent variable [16]. In Session 2, each participant performed two sets of half squat for each of the three loading conditions: optimal, or OL, as determined in Session 1, low, or LL (15% below OL), and high, or HL (15% above OL); totaling six sets. Each set was performed to failure or as a maximum of 20 repetitions. All participants performed the half squat using these loads in increasing order and all repetitions were executed as quickly as possible. Continuous repetition (CR) and 6 s IRR sets were performed in a counterbalanced order between participants. During each repetition, mean power (P_{mean}) and peak power (P_{peak}) were recorded by a linear velocity transducer. A summary of the experimental procedure for a representative subject of the study sample is depicted in Fig. 1.

2.3. Preliminary testing (Session 1)

Participants arrived at the laboratory having refrained from strenuous exercise for a minimum of 48 hours. Upon arrival, their height (Seca 202, Seca Ltd., Hamburg, Germany) and body mass (Tanita BC

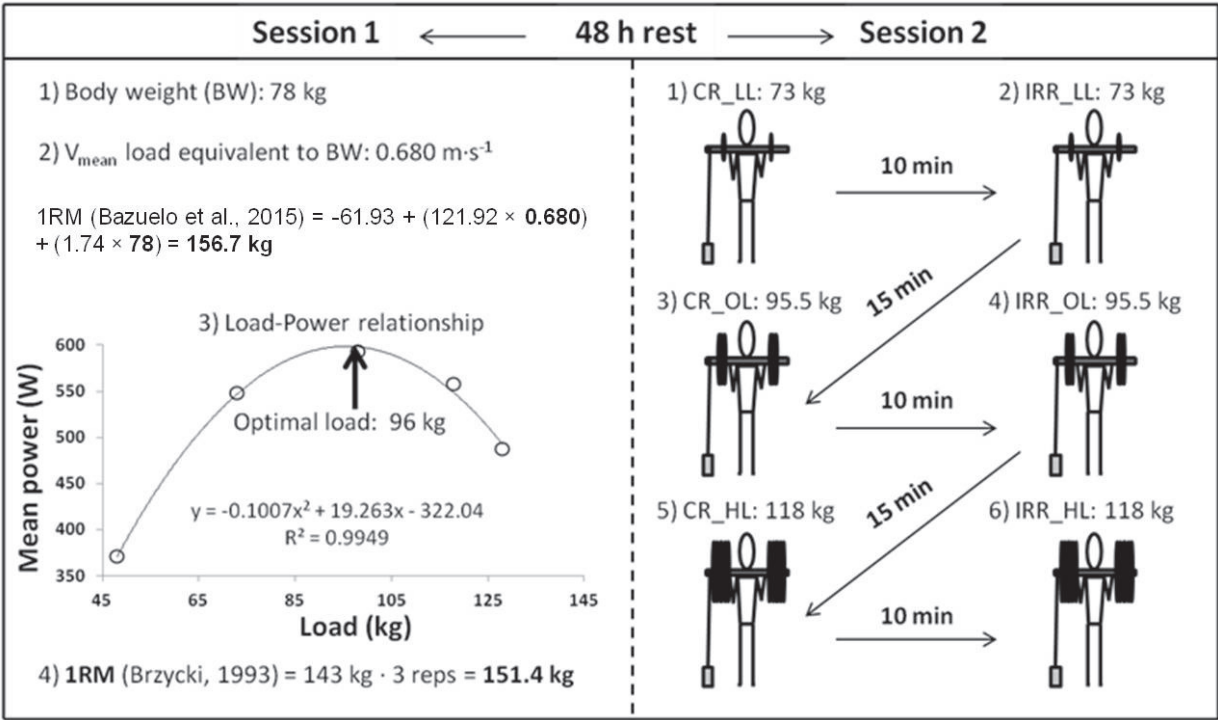


Fig. 1. Overview of the experimental protocol for a representative study participant. Note that continuous repetition (CR) and inter-repetition rest (IRR) sets were performed in random order counterbalanced between subjects.

418 segmental, Tokyo, Japan) were determined. Participants then completed a standard warm-up involving jogging, joint mobility, stretching exercises, and sub-maximal jump squats. At the end of the warm-up, each participant performed two sets of six and four repetitions (3 min of rest) as fast as possible in the half squat, lifting loads of 50% and 100% body weight (BW), respectively. The highest mean velocity (V_{mean}) recorded during the 100% BW-load repetition was used to estimate the individual's 1RM using the following equation: $1RM = -61.93 + (121.92 \cdot V_{mean}) + (1.74 \cdot LoadBW)$ [17]. This equation was only used to determine the loads for Session 1.

Individual load-power relationships were established in a progressive loading test. Participants performed 3 initial loads with 30%, 45% and 60% of the previously estimated 1RM. Afterwards, the load was gradually increased in 10% steps (70%RM, 80%RM, etc.) until a drop in P_{mean} was produced for two consecutive loads. Using each load, participants performed a single set of 2–3 repetitions. The passive recovery time between sets was 3 min for velocities $\geq 0.8 \text{ m} \cdot \text{s}^{-1}$ and 5 min for velocities $< 0.8 \text{ m} \cdot \text{s}^{-1}$. Finally, 1RM was estimated using the Brzycki equation [18] at a maximum of six repetitions. The 1RM obtained by

the Brzycki equation was used to determine the loads for Session 2.

For the half squat, the participant first stood with knees and hips fully extended, feet approximately shoulder-width apart, and the barbell resting across the back at the level of the acromion. The descent was executed as a continuous 2 s movement until a knee angle of 90° set by a manual goniometer, and this position was maintained for 1 s (velocity = $0 \text{ m} \cdot \text{s}^{-1}$). From this position, the participant was instructed to perform a purely concentric action (as fast as possible) to return to the starting position. The duration of eccentric and isometric phases was paced by auditory signals through an ad hoc audio file, while the concentric phase was always performed explosively at the maximum possible speed. All tests were performed with a Smith machine.

A dynamic measurement system (T-Force System; Ergotech, Murcia, Spain) was fixed perpendicularly to the bar with a tether to directly record its vertical instantaneous velocity at a frequency of 1000 Hz. Mean power (P_{mean}) and V_{mean} were defined as the area under the concentric portion of their respective curves divided by the duration of the repetition, whereas P_{peak} was determined as the maximum instantaneous value

achieved during the concentric phase at a given load. Only the best repetition executed at each load, identified according to the criterion of the highest P_{mean} , was considered for subsequent analysis. Optimal load was defined as the load that maximizes P_{mean} and was predicted by applying a polynomial equation to the individual load- P_{mean} curve [16].

2.4. Protocol testing (Session 2)

All participants arrived at the laboratory 48–72 hours after completing Session 1. In Session 2, they were required to undertake six sets (two per load) in increasing order of load (LL – OL – HL). The half squat movement pattern was the same as described for the first session. For each load, one set was performed without rest (CR) and the other with 6 s of rest between repetitions (IRR). During the IRR periods the barbell was supported on the stanchions of the Smith machine. The order of the CR and IRR sets using a given load was counterbalanced between participants.

All sets were performed to failure restricting the maximum number of repetitions to 20. Thus, a set ended when the participant was unable to perform another repetition, unable to follow the guidelines set by the audio file or had completed 20 repetitions. The recovery time between sets of the same load was 10 min and 15 min between sets of different loads.

2.5. Statistical analysis

Data are presented as the mean $\pm$ standard deviation (SD). A 2-way repeated measures ANOVA (set configuration [CR vs IRR] $\times$ load [low vs optimal vs heavy]) was applied to assess the differences in the number of repetitions performed. Decreases in P_{mean} and P_{peak} were assessed by 1-way repeated measures ANOVA. When the ANOVAs were significant, pairwise differences between means were identified using Bonferroni post hoc procedures. Within the same protocol, power outputs of the two initial repetitions were compared using a t-test for paired data. The magnitude of the differences was expressed as a standardized mean difference (Cohen's d effect size; ES). The ES was interpreted as: < 0.2 = trivial, 0.2 – 0.6 = small, 0.6 – 1.2 = moderate, 1.2 – 2.0 = large, and > 2 = very large [19]. Only data for the first 12, 9, and 6 repetitions performed at LL, OL, and HL, respectively, were entered in the analysis. All statistical tests were performed using the software package SPSS version 20.0 (SPSS, Chicago, Illinois). Significance was set at $p < 0.05$.

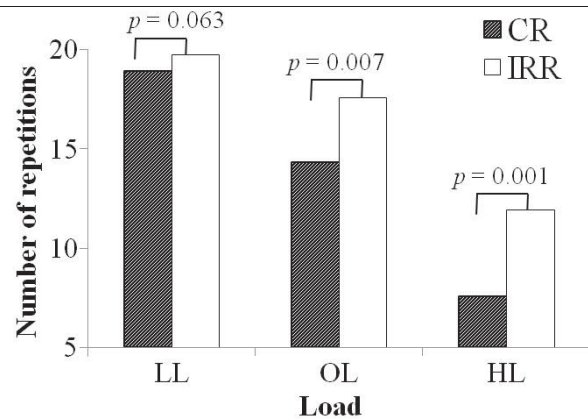


Fig. 2. Comparing the numbers of repetitions completed in CR (continuous repetition) and IRR (inter-repetition rest) protocols. LL, load below that linked to maximum power; OL, load linked to maximum power; HL, load above that linked to maximum power.

3. Results

The maximal dynamic strength (1RM) corresponded to 151.3 ± 19.5 kg obtained through a test of 3.6 ± 1.9 repetitions. The loads displaced were: 77.8 ± 11.8 kg LL, 101.1 ± 14.0 kg OL, and 123.4 ± 16.8 kg HL, corresponding to 51.5 ± 5.3 , 66.9 ± 5.4 and $81.6 \pm 5.2\%$ of 1RM respectively. The V_{mean} values recorded in the first repetition for LL, OL and HL, respectively, were: 0.591 ± 0.061 m $\cdot$ s $^{-1}$, 0.502 ± 0.061 m $\cdot$ s $^{-1}$, and 0.414 ± 0.055 m $\cdot$ s $^{-1}$.

The 2-way repeated measures ANOVAs conducted on the total number of repetitions performed revealed significant main effects either for set configuration ($F = 83.1$, $P < 0.001$, $\eta^2 = 0.922$), load ($F = 36.7$, $P < 0.001$, $\eta^2 = 0.710$), and the interaction set configuration $\times$ load ($F = 4.6$, $P = 0.029$, $\eta^2 = 0.397$). A higher number of repetitions per set were completed in the IRR compared to the CR protocol for LL (CR: 18.9 ± 2.3 vs. IRR: 19.8 ± 0.8 ; ES = 0.50), OL (CR: 14.3 ± 5.2 vs. IRR: 17.6 ± 4.2 ; ES = 0.66), and HL (CR: 7.6 ± 4.5 vs. IRR: 11.9 ± 4.8 ; ES = 0.98) (Fig. 2).

Power output losses, expressed as a percentage of the best power output recorded in the two initial repetitions ($Best2_{reps}$), produced in the six individual sets are depicted in Fig. 3. This plot shows a steeper decrease in power output when there was no rest between repetitions. During the CR protocol, significant relative P_{mean} reductions compared to the $Best2_{reps}$ were recorded in repetitions 7 ($p = 0.004$; ES = -1.39), 4 ($p = 0.002$; ES = -1.58), and 3 ($p = 0.012$; ES = -1.43) for LL, OL, and HL, respectively. Similarly, significant relative P_{peak} reductions versus $Best2_{reps}$

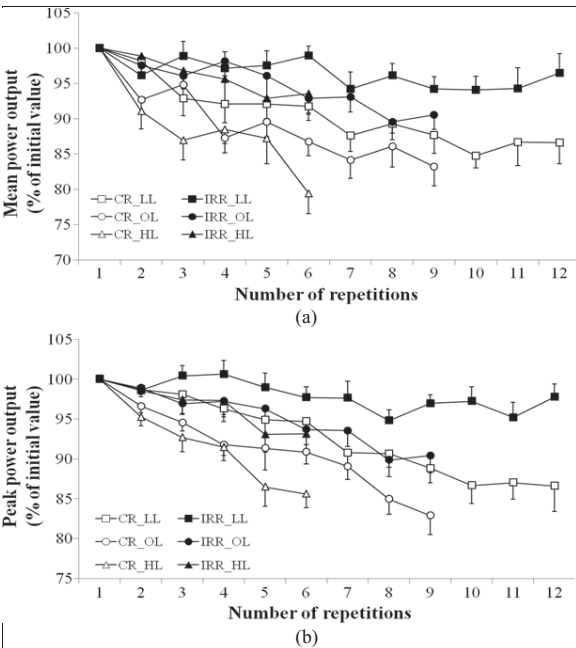


Fig. 3. Power output changes produced during the course of submaximal individual sets using three different loads. CR, continuous repetition protocol; IRR, inter-repetition rest protocol; LL, load below that linked to maximum power; OL, load linked to maximum power; HL, load above that linked to maximum power. (a) Mean power. (b) Peak power.

were detected in repetitions 7 ($p = 0.002$; $ES = -1.46$), 3 ($p = 0.007$; $ES = -1.38$), and 3 ($p = 0.034$; $ES = -1.23$) for LL, OL, and HL, respectively. In contrast, during the IRR protocol, significant P_{mean} and P_{peak} reductions were detected in repetition 8 for OL (P_{mean} : $p < 0.001$; $ES = -1.70$ and P_{peak} : $p < 0.001$; $ES = -1.25$), while no significant differences were observed for LL (P_{mean} : $p = 1.00$; $ES = -0.33$ and P_{peak} : $p = 1.00$; $ES = -0.35$ in repetition 12) and HL (P_{mean} : $p = 0.667$; $ES = -0.57$ and P_{peak} : $p = 0.147$; $ES = -0.77$ in repetition 6).

Table 1 shows the range of repetitions that could be performed based on two arbitrary percentage P_{mean} losses fixed at 8% and 15% of the $Best2_{reps}$. The IRR protocol allowed participants to double the number of repetitions per set for the three loads used while guaranteeing optimal power output. The $Best2_{reps}$ in terms of P_{mean} and P_{peak} was greater for the IRR protocol using the three loads (Table 2).

4. Discussion

The present study examined the acute effects of IRR to maintain lower limb muscular power when lifting

Table 1
Range of repetitions to perform based on the percentage of power loss than subjects are willing to assume

% Power loss	LL		OL		HL	
	CR	IRR	CR	IRR	CR	IRR
< 8%	1–5	1–17	1–3	1–7	1–2	1–6
8–15%	6–11	18–35	4–7	8–13	3–4	7–11
> 15%	12	36	8	14	5	12

% Power loss, mean power loss relative to the best value recorded in the two initial repetitions; LL, load below that linked to maximum power; OL, load linked to maximum power; HL, load above that linked to maximum power. CR, continuous repetition protocol; IRR, inter-repetition rest protocol.

loads commonly used during power training. Our results suggest by incorporating 6 s IRR periods in a half squat training protocol, power output is maintained across a larger number of repetitions compared with a similar protocol with no rest between repetitions. Several studies showed different CS configurations may attenuate the drop in power and velocity invariably produced by CR training [4,7,8]. Although periods of 15–30 s of IRR are commonly recommended for a CS [11], our results indicate 6 s rest periods could be sufficient to induce partial recovery in participants. Besides having beneficial impacts on muscle power development, short rest periods between repetitions could minimize the negative effects frequently associated with longer IRR periods, such as reduced gains in strength and muscle mass (hypertrophy), due to reduced post-exercise metabolic stress [4,20].

Ballistic exercises (e.g., jump squat) are generally preferred to traditional resistance training exercises (e.g., half squat) since the former are able to generate greater power output [12,21]. However, differences between both types of exercises tend to disappear as the load increases [22]. Accordingly, Requena et al. [23] recommend training in the traditional half squat within a range of loads close to the optimal load ($\sim 60\%$ 1RM; $75\% \sim 115\%$ of BW), as the power outputs elicited by these external loads show good correlation with performance in different functional tests. The OL recorded in our study (66.9% 1RM) was higher than the value described for athletes practicing different sports [24]. In addition, the V_{mean} eliciting maximal power was lower ($0.50 \text{ m} \cdot \text{s}^{-1}$) than the velocity reported by Izquierdo et al. [24] ($\sim 0.75 \text{ m} \cdot \text{s}^{-1}$). In contrast, the maximal dynamic strength observed for our study participants (1RM : 151.3 kg ; $1\text{RM} \cdot \text{BW}^{-1}$: $1.96 \text{ kg} \cdot \text{kg}^{-1}$) was similar to that provided for the group of athletes examined by Izquierdo et al. [24] (1RM : 154.5 kg ; $1\text{RM} \cdot \text{BW}^{-1}$: $2.08 \text{ kg} \cdot \text{kg}^{-1}$). Our results indicate the inability of the participants tested to express large

Table 2
Comparison of the power output in the first 2 repetitions

Variable	Load	Protocol	Repetition 1 (W)	Repetition 2 (W)	Best2 _{reps} (W)
Mean power	LL	CR	444.9 ± 70.1	446.8 ± 79.2	454.3 ± 75.1
		IRR	462.9 ± 95.1	447.0 ± 87.7	476.4 ± 88.7*
	OL	CR	487.9 ± 89.2	462.2 ± 102.7	490.7 ± 88.1
		IRR	505.7 ± 93.3	506.9 ± 85.3*	520.5 ± 94.9*
	HL	CR	502.2 ± 86.6	466.6 ± 85.1	507.7 ± 85.2
		IRR	484.3 ± 77.9	515.1 ± 105.0*	520.1 ± 102.3
Peak Power	LL	CR	1153.6 ± 232.1	1150.2 ± 207.3	1167.0 ± 219.7
		IRR	1178.0 ± 277.5	1213.6 ± 247.5*	1230.2 ± 254.2*
	OL	CR	1266.1 ± 241.4	1232.7 ± 241.4	1271.8 ± 236.0
		IRR	1360.1 ± 286.5*	1372.4 ± 243.1*	1390.7 ± 269.8*
	HL	CR	1348.3 ± 220.2	1286.1 ± 222.6	1352.1 ± 215.9
		IRR	1323.3 ± 268.9	1348.9 ± 261.6*	1370.4 ± 284.1

LL, Load under maximum power; OL, Load linked to maximum power; HL, Load upper maximum power; CR, Continuous repetitions; IRR, Inter-repetition rest; Best2_{reps}, highest power output recorded in the two initial repetitions; W, Watts; *, Significantly greater than CR value ($p < 0.05$).

forces when light loads are lifted. This inability is possibly the outcome of their usual training routine oriented to hypertrophy (multiple sets of 6–12 repetitions per set performed at 60–80% 1RM and a slow speed of movement).

It is well known mechanical and metabolic stimuli play an important role in the development of muscular strength, power and hypertrophic adaptations [20]. Although CS training regimens enhance power adaptations [25,26], the lower metabolic stress induced by this kind of exercise was identified as a limiting factor for the development of muscle strength and hypertrophy [4]. The novelty of our study was the use of shorter rest periods between repetitions produced a significant increase in the exercise volume accomplished for each load used. However, a limitation of our study was participants were instructed to stop when they had completed 20 repetitions, making the comparison of the number of repetitions not entirely accurate. We speculate that greater differences in the number of repetitions between IRR and CR would have been observed if the participants had continued performing repetitions until failure.

Izquierdo et al. [6] determined the optimal number of repetitions in the parallel back squat required to ensure maximal performance velocity maintenance working with loads corresponding to different percentages of maximal dynamic strength (60, 65, 70, and 75% of 1RM). Significant drops in average relative velocity were observed in repetitions 5, 9, 11, and 15 during the sets performed at 75%, 70%, 65%, and 60% 1RM, respectively. In the present study, earlier significant losses in average relative power output were produced in the absence of rest periods between repetitions (repetitions 7, 4 and 3 for LL [$\sim 51.5 \pm 5.3\%$

1RM], OL [$\sim 66.9 \pm 5.4\%$ 1RM], and HL [$\sim 81.6 \pm 5.2\%$ 1RM], respectively). The earlier onset of power output losses observed in our study may in part be explained by the better physical condition of the athletes participating in the study by Izquierdo et al. [6] (members of the Spanish national Basque ball team). Nevertheless, our results are in agreement with those of Baker and Newton [5], who suggested that power output drops quickly after 5 to 6 repetitions. As expected, when short rest periods (6 s) were introduced between repetitions, it took longer for a significant power output loss to occur.

Given the interest in avoiding muscle fatigue during power training, some authors established thresholds to determine when a set should be terminated to ensure minimal fatigue with training. For example, McBride et al. [1] ended training sets when power output decreased to under the arbitrary level of 15%, as this value corresponded to a significant decrease in barbell velocity. Similarly, Izquierdo et al. [6] reported significant velocity losses close to 8% of the velocity reached during the two initial repetitions. When resting 6 s between repetitions as opposed to no rest, setting a P_{mean} loss threshold of 8% led to the participants tested being able to increase the number of repetitions per set by 240% (5 vs. 17 repetitions for LL), 133% (3 vs. 7 repetitions for OL), and 200% (2 vs. 6 repetitions for HL). Accordingly, when this threshold was set at 15%, the corresponding increases in the number of repetitions per set performed were 218% (11 vs. 35 repetitions for LL), 86% (7 vs. 13 repetitions for OL), and 175% (4 vs. 11 repetitions for HL). We therefore propose incorporation of 6 s IRR periods during sets of half squat exercise will allow practitioners to double the volume of training sets using the three loads established while maintaining optimal power output.

Finally, we should mention that to compare power losses between the two protocols we considered the best power values recorded in the first two repetitions ($\text{Best}_{2\text{reps}}$). This is important because P_{mean} and P_{peak} values were always greater for the IRR protocol. Hence, absolute P_{mean} and P_{peak} were always higher for a similar relative power output loss in the IRR protocol, highlighting the differences between the two protocols.

5. Conclusions

The results of the present study confirm the usefulness of short IRR periods for the maintenance of power output during high repetition training sets. In fact, 6 s IRR serve to increase the total volume of training (more repetitions per set) and the quality of the repetitions within a set (higher power output). Thus, coaches should consider IRR periods when designing training programs for the development of muscular power. Short IRR periods may be desirable for athletes not wishing to reduce the stimulus on strength and muscle mass when trying to improve muscular power.

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Conflict of interest

The authors declare no conflicts of interest.

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